

Basic Principles of Medicine 2

Module: Neuroscience and Behaviour

Clinical Neuroimaging:

Directed Learning Activity

To be studied after Lecture

Organization of the Nervous

System

Images from Neuroanatomy: An Atlas of Structures, Sections and Systems 10th Ed. D.E. Haines © 2019 Lippincott Williams and Wilkins unless otherwise stated

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Recommended Reading

1. Siegel and Sapru, Essential Neuroscience, Fourth Edition, 2019, Chapter 27, Vascular Syndromes.

Part I: X-Ray

Learning Objectives

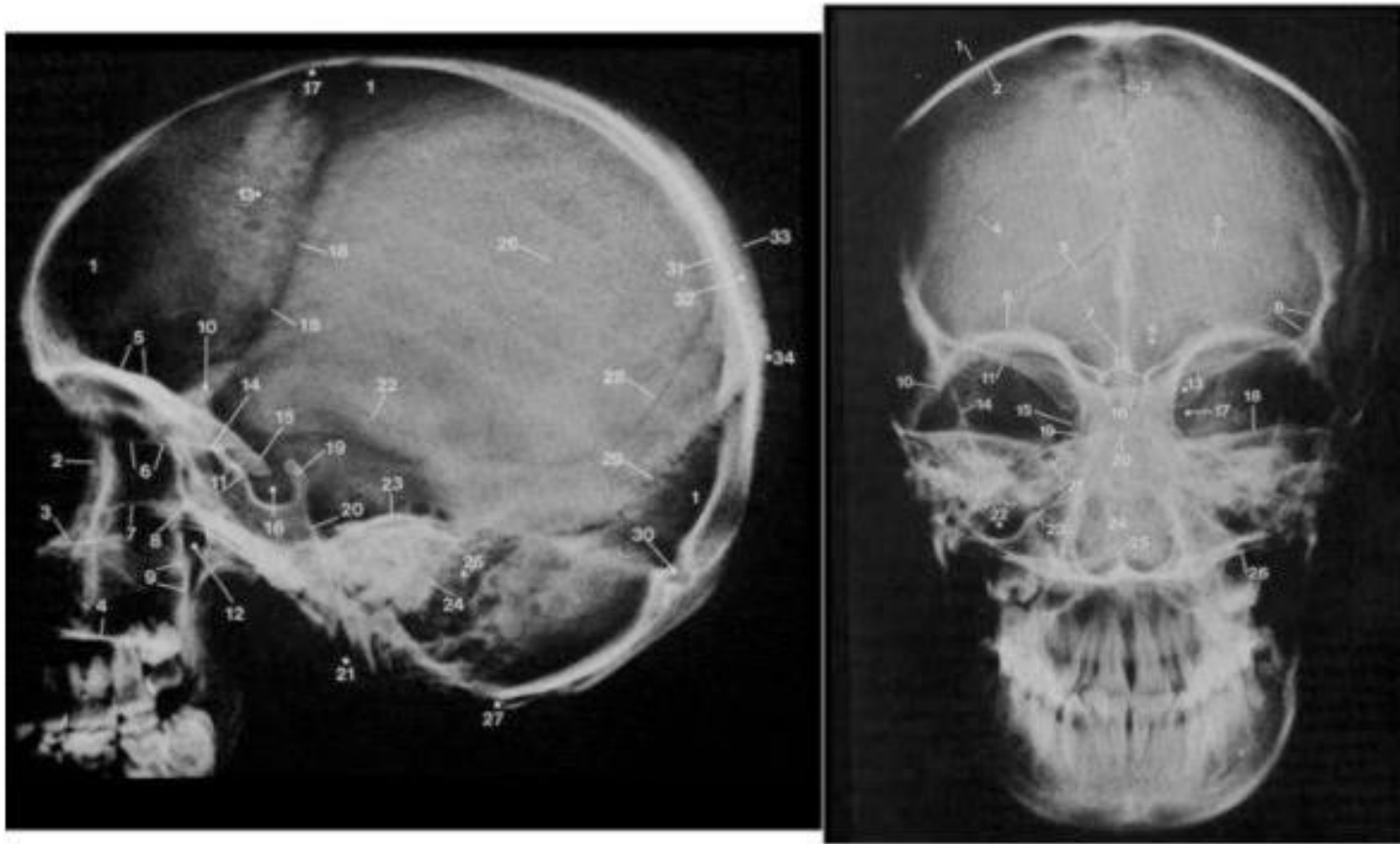
SOM.CS.I.BPM2.1.NSCI.0048 Describe the physics, diagnostic use and contraindications/limitations of X-ray-based techniques (excluding CT).

SOM.CS.I.BPM2.1.NSCI.0049 Describe the physics, diagnostic use and contraindications/limitations of CT.

SOM.CS.I.BPM2.1.NSCI.0050 Describe the physics, diagnostic use and contraindications/limitations of MRI.

- A broad beam of X-rays penetrate the body. Tissues absorb x-rays as a function of density. Unabsorbed x-rays undergo detection for representation as a negative image. X-rays of the head are most useful for assessing diseases of the skull. Only indirect inference can therefore be drawn regarding pathologies of low-density tissue. For example, internal erosion of the skull may indicate an expanding neoplasm.

X-Ray Skull: Lateral and Frontal Views



Unabsorbed X-rays strike a detector (X-rays are sensitive to tissue density).
High-density tissues (e.g., bone) appear white

Nervous system is homogeneously low-density (nervous pathology revealed indirectly)

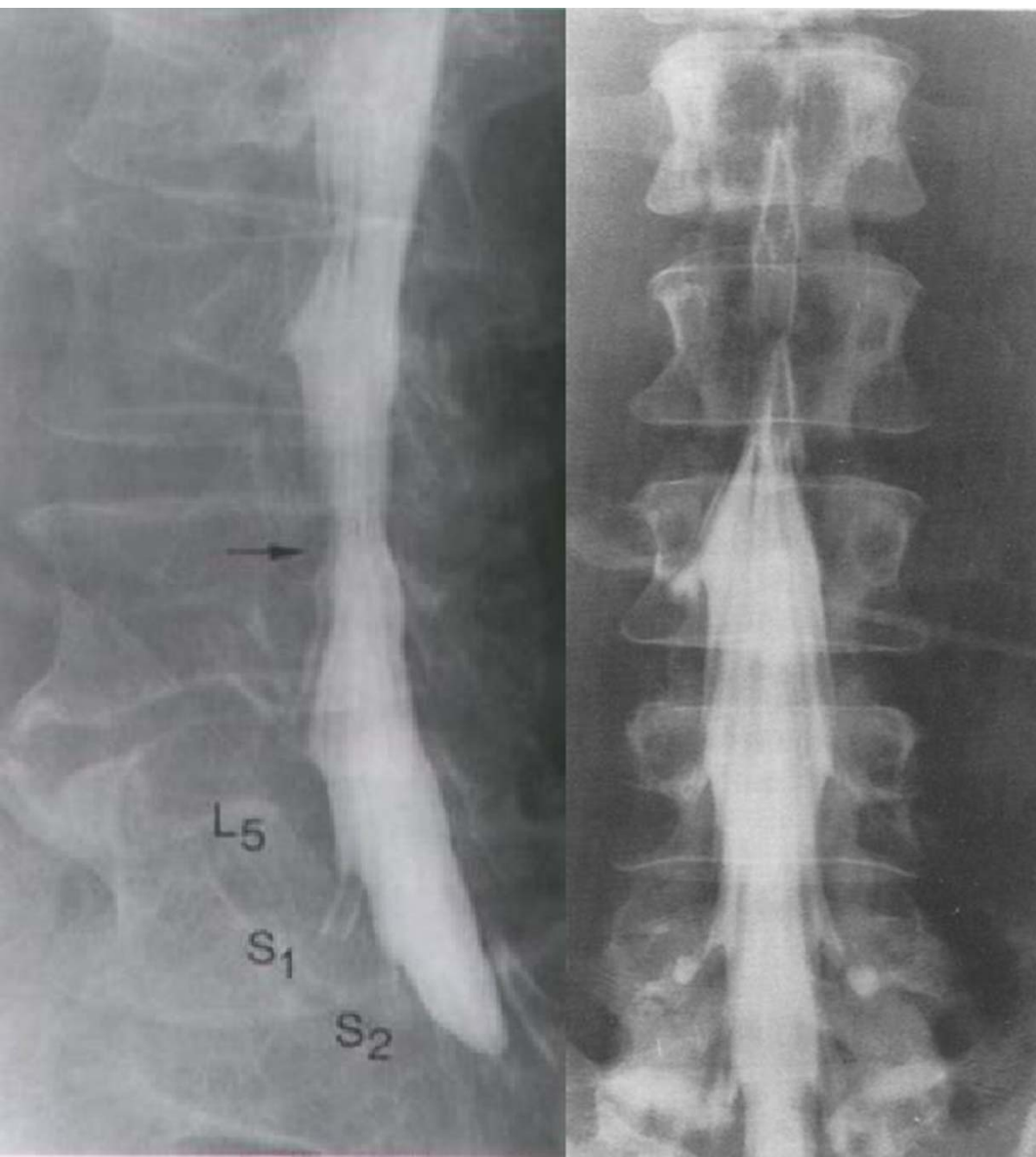
Masuhr, Neumann, Neurologie, MLP

X-ray Advantages and Disadvantages

- Advantages
 - Inexpensive
 - Commonly available
 - Sensitive to skull fracture and dense foreign matter
- Disadvantages
 - Exposure to ionizing radiation
 - Brain structures minimally visible



- In myelography, the spinal cord, nerve roots and the subarachnoid space are examined. The myelogram is taken after injecting a contrast medium into the spinal subarachnoid space. Myelography reveals herniated (slipped) intervertebral disks and spinal tumors.



- Contrast agent administered via lumbar puncture
- Irregularities in the subarachnoid space may reveal tumor or disk anomalies

Masuhr, Neumann, Neurologie, MLP, 1992

Myelography

Cerebral Angiography (Vasography)

- **Definition:**

- X-ray representation of blood vessels with contrast medium applied through a catheter

- **Risks:**

- Allergic reaction
- Arterial spasm → cerebral infarction

- **Indications:**

- Vascular malformations: Angioma (vascular tumor), aneurysm
- Vascular obstruction, stenosis

Cerebral Angiography (Vasography)

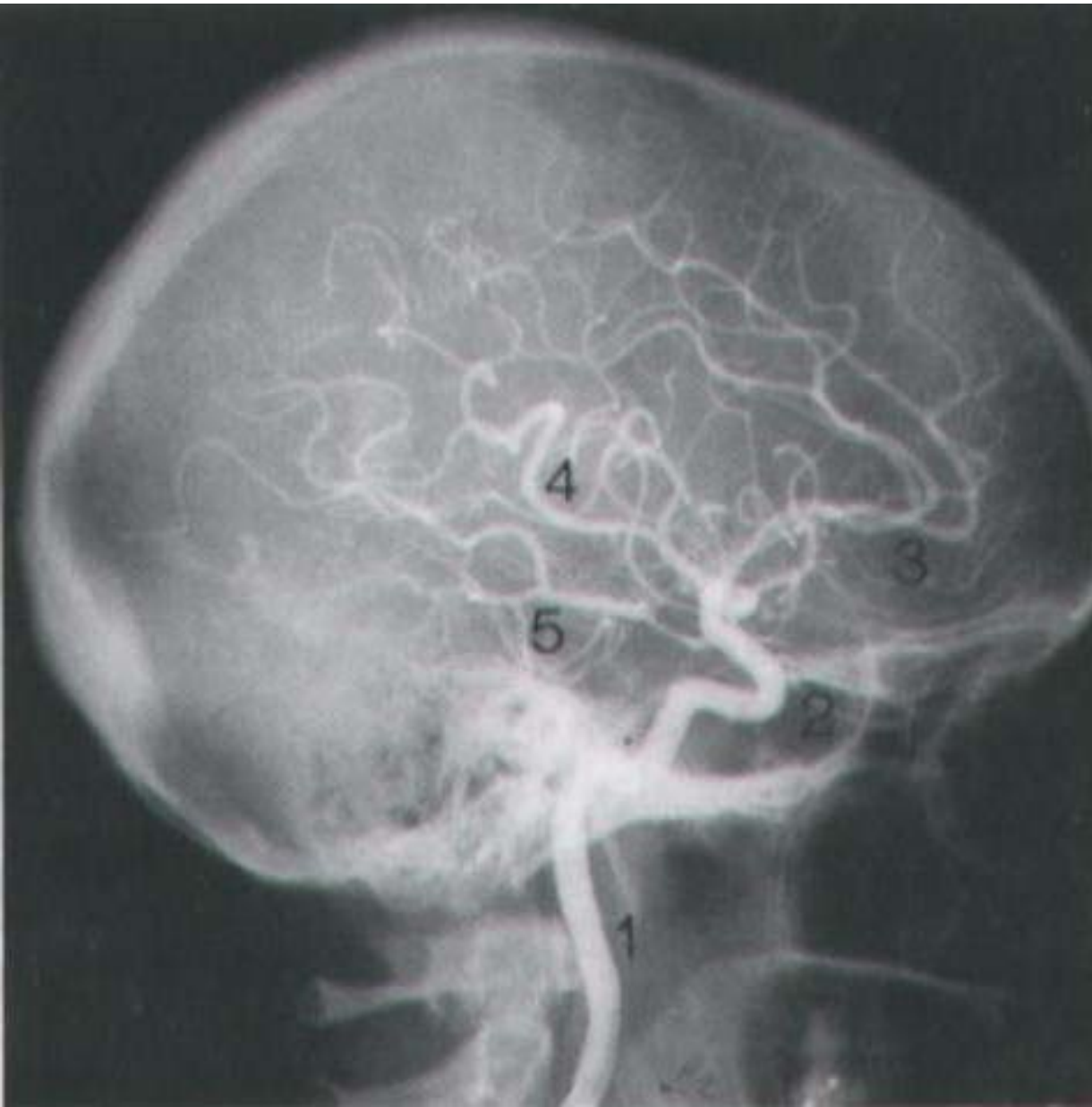
1. internal carotid artery
2. carotid siphon
3. anterior cerebral artery
4. medial cerebral artery
5. posterior cerebral artery

Radiographs taken as a radio-opaque contrast agent courses through the vessels.

Frontal View

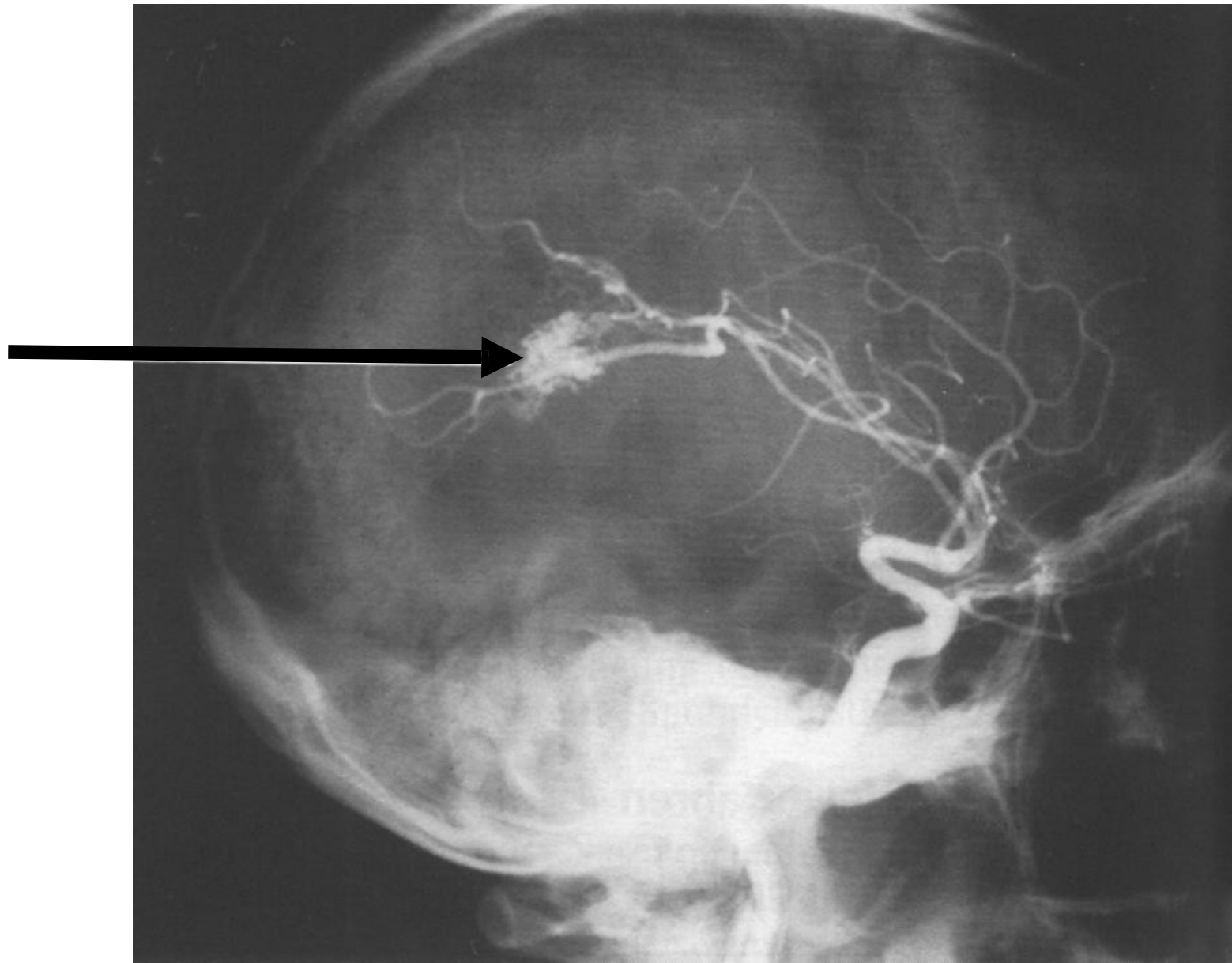


Cerebral Angiography (Vasography)



1. internal carotid artery
2. carotid siphon
3. anterior cerebral artery
4. medial cerebral artery
5. posterior cerebral artery

Lateral View



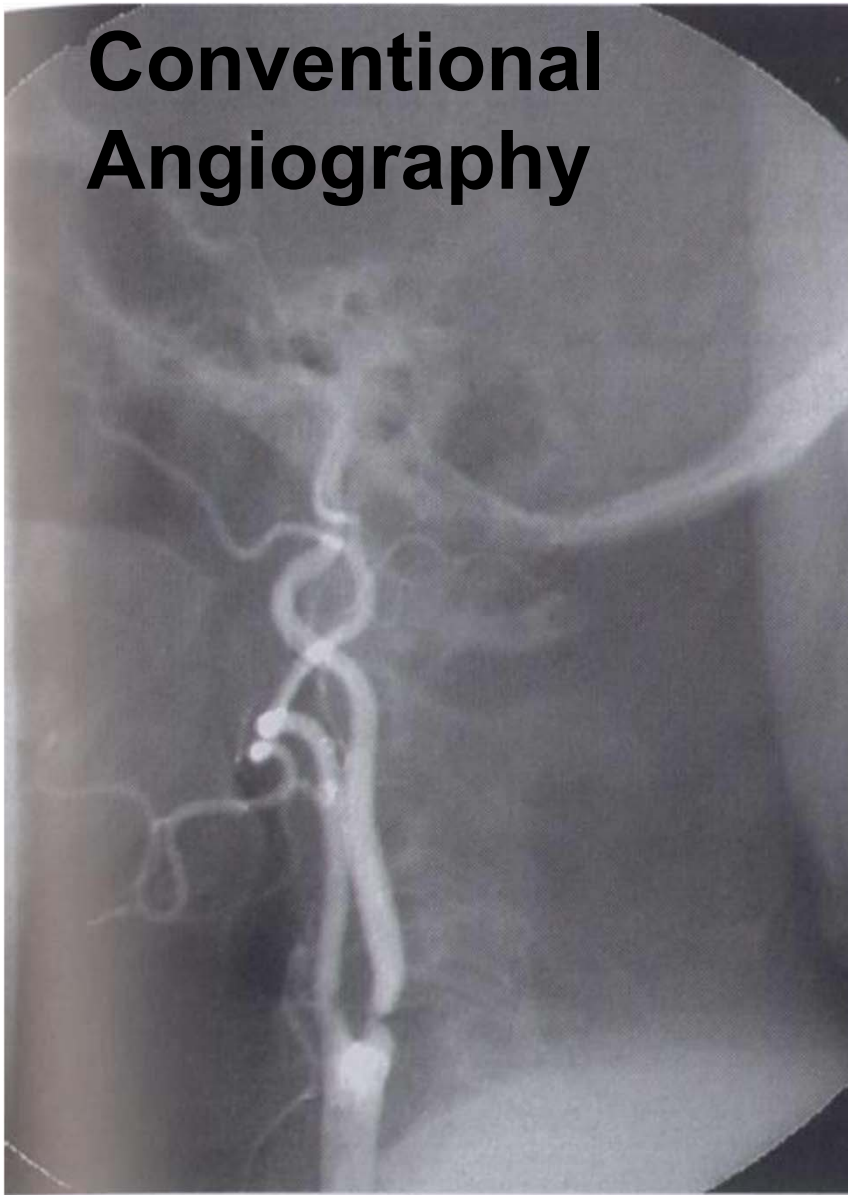
Masuhr, Neumann, Neurologie, MLP, 1992

X-ray angiograph showing middle cerebral arterio-venous malformation.

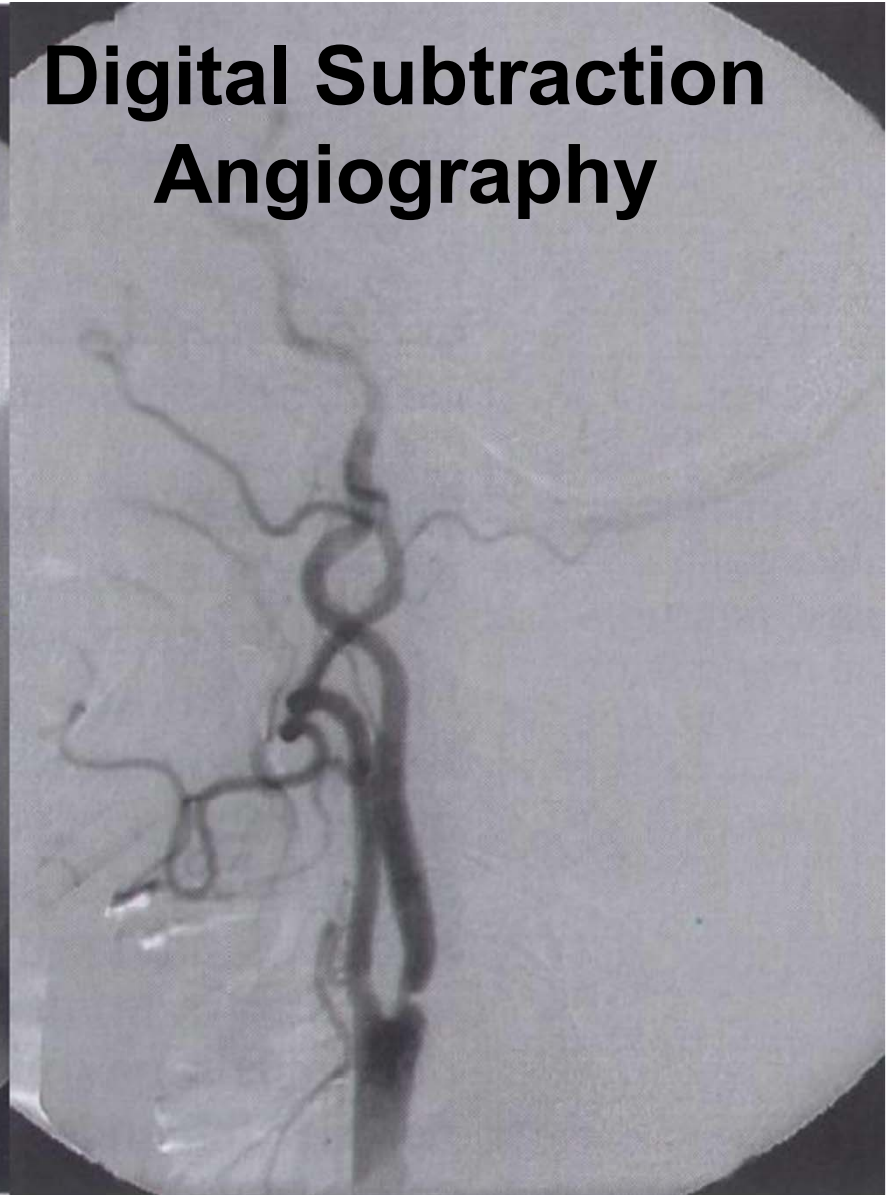
Digital Subtraction Angiography



Conventional Angiography

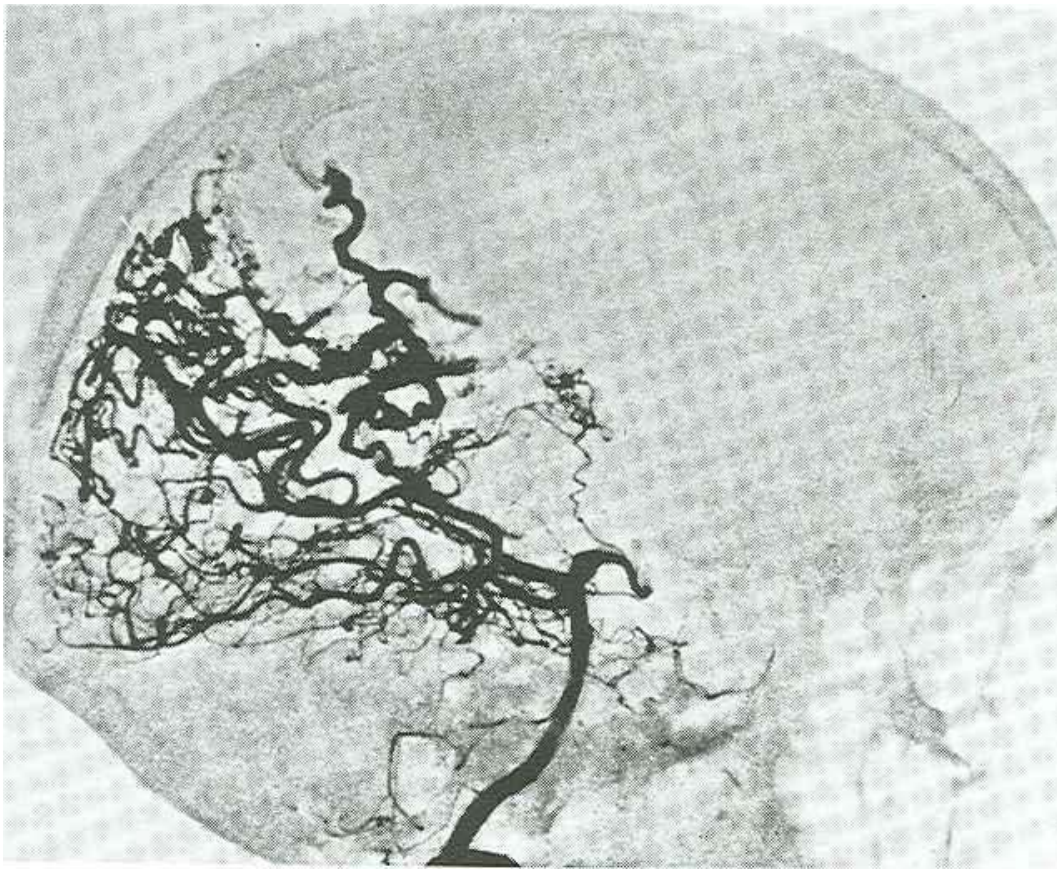


Digital Subtraction Angiography



- Native X-ray is taken followed by conventional angiograph
- The angiograph is digitally subtracted from the native X-ray, yielding a dark image of the vessels with attenuation of dense structures (e.g., bone)

DSA Angioma



Masuhr, Neumann, Neurologie, MLP, 1992